An overlapping-generations model for PolicyEngine reform scoring

OG-Core, the OG-UK calibration, and microdata-estimated tax functions[*]

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Abstract

This paper documents the overlapping-generations (OLG) member of the PolicyEngine Macro model suite: the open-source OG-Core dynamic general-equilibrium framework of DeBacker and Evans (2024a), deployed today through its United Kingdom country calibration OG-UK, with the United States calibration OG-USA in scope. The model tracks 80 annual age cohorts of heterogeneous lifetime-ability households who choose labour supply and saving over the life cycle, a perfectly competitive production sector, and a government that taxes, transfers, spends, and issues debt, and it solves both the long-run stationary steady state and the year-by-year transition path between steady states. Its distinguishing feature within the suite is the microdata bridge of DeBacker et al. (2019): statutory reforms are specified as PolicyEngine tax–benefit parameters, run through the PolicyEngine UK microsimulation model over enhanced Family Resources Survey microdata, and summarised as smooth parametric effective- and marginal-tax-rate functions that the optimising households respond to — so the general-equilibrium model sees the true shape of a reform across the income distribution rather than a stylised wedge. I set out the model's equations, the tax-function estimation pipeline, the UK calibration and how selected targets compare with official ONS and OBR figures, and the `policyengine-macro` command-line integration (`pe-macro score --model og`). I am explicit about status: the wiring is experimental-grade, scoring is steady-state-only in the CLI today, a single solve takes upwards of seventeen minutes so the model is excluded from the hosted service, and — unlike the suite's OBR and SVAR members — there is no published ground truth to replicate: the model delivers structural counterfactuals, not validated forecasts.

Keywords: overlapping generations, dynamic general equilibrium, fiscal policy, microsimulation, tax functions, dynamic scoring

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1 Introduction

PolicyEngine Macro is a suite of open-source macroeconomic models built around a single reform vocabulary: a tax–benefit reform is a flat dictionary of PolicyEngine parameter changes, and every member of the suite consumes that same object through its own declared contract. The suite's other members are anchored to published figures — an emulator of the Office for Budget Responsibility's (OBR) macroeconometric model, a Bank-of-England-style structural VAR, and the PolicyEngine static microsimulation itself. This paper documents the member that answers a different kind of question: the *psl-og* model, a dynamic overlapping-generations (OLG) general-equilibrium model built on the Policy Simulation Library's OG-Core framework (DeBacker and Evans, 2024a), wired into the suite today through its United Kingdom country calibration OG-UK (DeBacker et al., 2025), with the United States calibration OG-USA (DeBacker and Evans, 2024b) in scope.

The model is a large-scale descendant of Auerbach and Kotlikoff (1987). It tracks eighty annual age cohorts — households aged 20 to 99, entering economic life at 20 and facing age-specific mortality — each populated by several lifetime-ability types. Every cohort chooses labour supply and saving over its remaining lifetime, trading off consumption against an elliptical disutility of work and a warm-glow bequest motive. Perfectly competitive firms rent the resulting capital and labour; the government taxes, transfers, spends, and issues debt, with a fiscal rule that stabilises the long-run debt-to-GDP ratio. Two solution objects are on offer: the *stationary steady state* ("where does the economy settle under this policy?") and the *transition path* (a roughly 60-year, year-by-year solve of how it gets there).

Within the suite, the OLG member's role is long-run structural analysis. The OBR emulator and the SVAR speak to demand-side dynamics over a forecast window; the static microsimulation speaks to first-round revenue and distribution. The OLG model is the only member in which a tax reform changes prices — wages, interest rates — through households' labour-supply and saving responses, so it is the member that can address questions like the long-run output cost of a higher basic rate of income tax, the crowding-out effects of debt, or the incidence of corporation tax across generations. Its distinguishing input channel is the microdata bridge of DeBacker et al. (2019): reforms are not stylised wedges but actual statute, run through the PolicyEngine UK microsimulation model (Woodruff et al., 2025) over enhanced Family Resources Survey (FRS) microdata and compressed into smooth parametric effective- and marginal-tax-rate functions that the optimising households face.

Two properties of this member must be stated at the outset, because they shape everything downstream. First, it is *local-only*: a single steady-state solve — microdata calibration plus the general-equilibrium fixed point — has been measured at over seventeen minutes on a laptop, and a full transition path takes hours, so the model is deliberately excluded from the hosted PolicyEngine Macro service and runs only on the user's own machine through the `pe-macro` command-line interface. Second, unlike the suite's OBR and SVAR members, it is a *calibrated structural model with no published ground truth to replicate*. Its outputs are structural counterfactuals — internally consistent answers conditional on the model's assumptions — not validated forecasts. This paper is accordingly written as documentation with explicit caveats (Section 7), not as a validation exercise.

The remainder of the paper proceeds as follows. Section 2 places the model in the OLG literature and cites the framework's own documentation. Section 3 sets out the model's equations: the household problem, the firm problem, the government budget and fiscal rules, market clearing, and the steady-state and transition-path solution algorithms. Section 4 details the tax-function estimation pipeline from PolicyEngine microdata — the PolicyEngine-specific contribution. Section 5 describes the software architecture and the `policyengine-macro` integration. Section 6 documents the UK calibration and compares selected targets with official statistics. Section 7 states the model's limitations honestly, and Section 8 concludes.

2 Related literature and model provenance

The intellectual ancestry runs from the two-period consumption-loan and neoclassical OLG economies of Samuelson (1958) and Diamond (1965) to the 55-period computational life-cycle model of Auerbach and Kotlikoff (1987), which established the OLG model as the workhorse for quantitative fiscal-policy analysis — tax reform, social security, and generational incidence. Modern large-scale descendants add idiosyncratic ability heterogeneity, realistic demographics, and detailed fiscal institutions; Nishiyama and Smetters (2007) is a prominent example in the social-security literature, and the OBR has recently built its own UK OLG model in the same tradition (Office for Budget Responsibility, 2025b).

OG-Core is the open-source embodiment of this tradition maintained by the Policy Simulation Library community. Its theory, calibration interface, solution methods, and Python API are documented in DeBacker and Evans (2024a), which serves as the model's canonical reference; Evans and DeBacker (2024) provide a textbook treatment of building and solving OLG models for policy analysis that covers most of the structures in OG-Core, and Evans and Phillips (2014) analyse the time-path-iteration solution method the framework uses for transition dynamics. The framework separates *theory* (OG-Core, a country-agnostic system of equations and solvers) from *calibration* (country packages that supply demographics, tax detail, and macro parameters). OG-USA, the United States calibration, is documented in DeBacker and Evans (2024b) and has been used for dynamic scoring of US federal tax reform; OG-UK (DeBacker et al., 2025) is the United Kingdom calibration, originally built on OpenFisca-UK and now on PolicyEngine UK (Woodruff et al., 2025), sourcing calibration targets from the ONS, the OBR, the Bank of England, and GOV.UK.

The methodological bridge that makes the model usable for statutory reform scoring is DeBacker et al. (2019), who propose estimating parametric effective- and marginal-tax-rate functions — jointly in labour and capital income — from the output of a tax microsimulation model, and embedding those functions in the dynamic general-equilibrium model's household problem. The functional-form menu draws on the empirical tax-function literature: the three-parameter total-income schedule of Gouveia and Strauss (1994) and the log-linear progressivity form of Heathcote et al. (2017), alongside DeBacker–Evans–Phillips's own richer bivariate specification. PolicyEngine Macro's contribution is not to this theory but to the pipeline: replacing the microsimulation input with PolicyEngine's open-source UK model over enhanced FRS microdata, and exposing the whole loop — reform in, steady-state general-equilibrium impact out — behind the suite's common scoring interface.

3 The model

This section reproduces the core blocks of OG-Core following DeBacker and Evans (2024a), in the configuration used by the PolicyEngine Macro integration: a single production sector, and households differentiated by age and lifetime ability. Time is annual. Households live at most $E + S$ periods: E economically inactive childhood years (ages 0–19) and $S = 80$ active years (model ages $s = E+1,\dots,E+S$, real ages 20–99). Each cohort contains J lifetime-ability types indexed by j, with population shares λ_j ($\sum_j \lambda_j = 1$) and deterministic age–ability effective-labour profiles $e_{j,s}$. Population $\omega_{s,t}$ evolves by exogenous fertility, mortality ρ_s, and immigration; labour-augmenting productivity grows at rate g_y, and all equations below are expressed before stationarisation (the solved system divides by $e^{g_y t}$ and population growth to obtain a stationary equivalent).

3.1 Households

A household of type j and age s at time t has period utility that is additively separable in consumption $c_{j,s,t}$, labour $n_{j,s,t}$, and bequests-to-be-left $b_{j,s+1,t+1}$:

$$u(c_{j,s,t}, n_{j,s,t}, b_{j,s+1,t+1}) \;=\; \frac{c_{j,s,t}^{1-\sigma}-1}{1-\sigma} \;+\; e^{g_y t(1-\sigma)}\,\chi^n_s\, v(n_{j,s,t}) \;+\; \chi^b_j\,\rho_s\,\frac{b_{j,s+1,t+1}^{1-\sigma}-1}{1-\sigma}, \tag{1}$$

where σ is the coefficient of relative risk aversion, χ^n_s is an age-specific disutility-of-labour scale, and the warm-glow bequest term is weighted by the mortality hazard ρ_s and an ability-specific intensity χ^b_j. The disutility of labour takes the *elliptical* form

$$v(n) \;=\; -\,b\left[1-\left(\frac{n}{\tilde{l}}\right)^{\upsilon}\right]^{1/\upsilon}, \qquad b,\upsilon > 0, \tag{2}$$

where $\tilde{l}$ is the time endowment. The ellipse parameters (b,υ) are fitted so that the implied labour-supply behaviour matches a constant-Frisch-elasticity specification with elasticity θ over the interior of the choice set, while — unlike the CFE form — delivering marginal disutility that goes to infinity at both $n = 0$ and $n = \tilde{l}$, so the upper and lower bounds on labour supply never bind and Kuhn–Tucker conditions are unnecessary (Evans and Phillips, 2014; DeBacker and Evans, 2024a).

The period budget constraint, in the single-good configuration used here (suppressing the multi-good consumption-tax detail of the general framework), is

$$(1+\tau^c_t)\,c_{j,s,t}+b_{j,s+1,t+1} \;=\; (1+r_{p,t})\,b_{j,s,t}+w_t\,e_{j,s}\,n_{j,s,t}+bq_{j,s,t}+tr_{j,s,t}+pension_{j,s,t}-tax_{j,s,t}, \tag{3}$$

where $b_{j,s,t}$ is beginning-of-period wealth, $r_{p,t}$ is the return on the household's portfolio of private capital and government bonds, w_t is the wage per effective labour unit, $bq_{j,s,t}$ are received bequests (accidental bequests are redistributed within ability type), $tr_{j,s,t}$ are government transfers, and $tax_{j,s,t} = \tau^{etr}(x_{j,s,t}, y_{j,s,t})\,(x_{j,s,t}+y_{j,s,t})$ is the net income-tax liability generated by the effective-tax-rate function of Section 4, evaluated at labour income $x_{j,s,t} \equiv w_t e_{j,s} n_{j,s,t}$ and capital income $y_{j,s,t} \equiv r_{p,t} b_{j,s,t}$.

The household chooses $\{n_{j,s,t}, b_{j,s+1,t+1}\}$ to maximise expected discounted lifetime utility with discount factor β and survival probabilities $(1-\rho_s)$. The labour-supply first-order condition equates the after-tax marginal return to work with the marginal disutility of labour:

$$w_t e_{j,s}\big(1-\tau^{mtrx}_{s,t}\big)\,(c_{j,s,t})^{-\sigma} \;=\; e^{g_y(1-\sigma)}\,\chi^n_s\,v'(n_{j,s,t}), \tag{4}$$

where $\tau^{mtrx}_{s,t}$ is the marginal tax rate on labour income implied by the estimated tax functions. The consumption–saving Euler equation for ages $s < E+S$ is

$$(c_{j,s,t})^{-\sigma} \;=\; \beta(1-\rho_s)\left[1+r_{p,t+1}\big(1-\tau^{mtry}_{s+1,t+1}\big)\right](c_{j,s+1,t+1})^{-\sigma} \;+\; \chi^b_j\,\rho_s\,(b_{j,s+1,t+1})^{-\sigma}, \tag{5}$$

with τ^{mtry} the marginal rate on capital income, and in the final period of life the household equates the marginal utility of consumption with the marginal warm-glow value of the terminal bequest, $(c_{j,E+S,t})^{-\sigma} = \chi^b_j (b_{j,E+S+1,t+1})^{-\sigma}$. Equations (4)–(5) show precisely where the microdata-estimated tax functions bite: reforms move τ^{etr}, τ^{mtrx} and τ^{mtry} across the joint income distribution, and the general equilibrium reprices w_t and r_t in response.

3.2 Firms

A unit measure of perfectly competitive firms produces the single output with constant-elasticity-of-substitution technology in private capital K_t and effective labour L_t (the deployed configuration switches off public-capital augmentation):

$$Y_t \;=\; F(K_t, L_t) \;=\; Z_t\Big[\gamma^{1/\varepsilon} K_t^{\frac{\varepsilon-1}{\varepsilon}} + (1-\gamma)^{1/\varepsilon}\big(e^{g_y t} L_t\big)^{\frac{\varepsilon-1}{\varepsilon}}\Big]^{\frac{\varepsilon}{\varepsilon-1}}, \tag{6}$$

where Z_t is total factor productivity, γ is the capital share parameter, and ε is the elasticity of substitution. In the Cobb–Douglas limit $\varepsilon = 1$, (6) reduces to $Y_t = Z_t K_t^{\gamma}\big(e^{g_y t} L_t\big)^{1-\gamma}$. Firms face a corporate income tax at rate τ^b_t with a deduction for depreciation at rate δ^τ_t (which may differ from economic depreciation δ), and choose K_t, L_t to maximise after-tax profits

$$PR_t \;=\; (1-\tau^b_t)\big(Y_t - w_t L_t\big) - \big(r_t + \delta\big) K_t + \tau^b_t\, \delta^\tau_t K_t. \tag{7}$$

The first-order conditions deliver the factor-price equations

$$r_t \;=\; (1-\tau^b_t)\,\frac{\partial F}{\partial K_t} - \delta + \tau^b_t\, \delta^\tau_t, \qquad w_t \;=\; \frac{\partial F}{\partial L_t}. \tag{8}$$

The corporation-tax rate τ^b is therefore a structural parameter of the macro model rather than a PolicyEngine parameter; in the integration it is shocked directly through parameter overrides (Section 5).

3.3 Government

The government levies household income taxes through the estimated tax functions, consumption taxes, and the corporate tax; it spends on public consumption G_t, non-pension transfers TR_t, and pensions; and it issues one-period debt D_t at rate $r_{gov,t}$ (a possibly wedge-adjusted function of r_t). The flow budget constraint is

$$D_{t+1} + Rev_t \;=\; (1 + r_{gov,t})\, D_t + G_t + TR_t + Pensions_t, \tag{9}$$

where Rev_t aggregates all tax receipts. Along the baseline, transfers follow the rule $TR_t = \alpha_{tr}\, Y_t$ and public consumption $G_t = \alpha_g\, Y_t$ as fixed GDP shares. Because (9) does not by itself pin down a stationary debt path, OG-Core imposes a *fiscal closure rule*: between periods T_{G1} and T_{G2} of the transition, G_t (or TR_t, or both, by configuration) adjusts endogenously so that the debt-to-GDP ratio converges to a long-run target α_D, and holds it there thereafter. In the steady state, debt is exactly $\bar{D} = \alpha_D \bar{Y}$ and (9) determines the residual spending level consistent with it.

3.4 Market clearing and equilibrium

Equilibrium requires three markets to clear. Labour supply aggregates to

$$L_t \;=\; \sum_{s=E+1}^{E+S} \sum_{j=1}^{J} \omega_{s,t}\, \lambda_j\, e_{j,s}\, n_{j,s,t}, \tag{10}$$

household savings fund private capital and government debt (in the closed-economy configuration; the deployed small-open-economy features allow foreign holdings of debt at an exogenous world rate),

$$K_t + D_t \;=\; \sum_{s=E+2}^{E+S+1} \sum_{j=1}^{J} \omega_{s-1,t-1}\, \lambda_j\, b_{j,s,t}, \tag{11}$$

and the goods market clears by Walras's law:

$$Y_t \;=\; C_t + I_t + G_t, \qquad I_t = K_{t+1} - (1-\delta)K_t. \tag{12}$$

A *steady-state equilibrium* is a stationary allocation and price pair $(\bar{r}, \bar{w})$ such that households optimise, firms optimise, the government satisfies its closure rule, and all markets clear with time-invariant stationarised quantities. A *non-steady-state equilibrium* is a full time path of allocations and prices from arbitrary initial conditions converging to that steady state.

3.5 Solution algorithms

Steady state. The steady state is computed as a fixed point in a small vector of aggregates. An outer loop guesses $(\bar{r}, \overline{BQ}, \overline{TR}, \text{factor})$ — the interest rate, aggregate bequests, aggregate transfers, and the model-to-data income scaling factor used by the tax functions. Given the guess, factor prices follow from (8); an inner loop then solves each ability-type's lifetime problem — the $2S$ nonlinear equations (4)–(5) — by a root finder, aggregates the resulting policies over (j, s), and produces implied aggregates. The outer loop updates the guess by damped iteration until the implied and guessed vectors agree to tolerance. The integration exposes the iteration cap (`max_iter`, default 250 in the CLI).

Transition path. Transition dynamics are solved by *time-path iteration* (TPI) in the tradition of Auerbach and Kotlikoff (1987), analysed for this class of models by Evans and Phillips (2014): guess entire time paths $\{r_t, w_t, BQ_t, TR_t\}_{t=1}^{T}$ that end at the new steady state; given those paths, solve every cohort's remaining-lifetime problem (including cohorts alive at the reform date, whose problems are truncated); aggregate to implied paths; and update the guessed paths by damped iteration until convergence. With $T \approx 320$ periods, $S = 80$ ages, and J ability types, one TPI solve is orders of magnitude more expensive than a steady-state solve; the implementation distributes the lifetime problems across Dask workers. In the PolicyEngine Macro CLI, only steady-state comparison is wired today; transition paths are available through the underlying `oguk` API.

4 Tax functions from PolicyEngine microdata

The channel through which a statutory reform enters the general-equilibrium model is the tax-function estimation step of DeBacker et al. (2019). This is the PolicyEngine-specific part of the pipeline, and it is what distinguishes the psl-og member from a stylised-wedge OLG exercise: the households in Section 3.1 respond to a smooth summary of the *actual* UK tax and benefit code, computed reform-by-reform from microdata.

4.1 The pipeline

The estimation loop, run once per budget-window year at calibration time (for both the baseline and each reform), is:

1. **Microsimulation.** PolicyEngine UK (Woodruff et al., 2025) is run over the PolicyEngine enhanced FRS microdata — the Family Resources Survey reweighted and enriched with HMRC Survey of Personal Incomes, Living Costs and Food Survey, and Wealth and Assets Survey information — under the policy in question. This yields, for every adult in the sample, total tax liability, labour income x, and capital income y, and hence an *effective tax rate* $ETR = tax/(x + y)$.

2. **Marginal rates by perturbation.** Marginal tax rates are computed numerically: add £1 of employment income (for τ^{mtrx}) or of capital income (for τ^{mtry}) per adult, rerun the microsimulation, and read off the change in household net income. This captures the full statutory schedule including benefit tapers, thresholds, and interactions — e.g. the Universal Credit taper and personal-allowance withdrawal appear in the marginal-rate cloud automatically.

3. **Function fitting.** For each year (and, if configured, each age group), parametric functions $\tau^{etr}(x, y)$, $\tau^{mtrx}(x, y)$, $\tau^{mtry}(x, y)$ are fitted to the population-weighted scatter of micro observations by nonlinear least squares (OG-Core's `txfunc` module).

4. **Embedding.** The fitted parameters are handed to OG-Core, where the functions appear in the households' budget constraints and Euler equations (3)–(5), scaled between model units and pounds by the endogenous *factor* solved alongside the steady state.

A reform score is then the comparison of two full general-equilibrium solves whose only primitive difference is the fitted tax-function parameters (plus any structural overrides).

4.2 Functional forms

OG-Core's default specification is the *DEP* (DeBacker–Evans–Phillips) form: a bivariate function of labour income x and capital income y built as a Cobb–Douglas aggregate, with weight $\phi \in [0, 1]$, of two univariate ratio-of-polynomials schedules,

$$\tau(x, y) \;=\; \Big[\tau^x(x) + shift_x\Big]^{\phi} \Big[\tau^y(y) + shift_y\Big]^{1-\phi} + shift, \tag{13}$$

where each univariate component is a bounded, monotone ratio of quadratics,

$$\tau^x(x) \;=\; \big(max_x - min_x\big)\,\frac{Ax^2 + Bx}{Ax^2 + Bx + 1} \;+\; min_x, \qquad A, B > 0, \tag{14}$$

and symmetrically for $\tau^y(y)$ with parameters (C, D, max_y, min_y). The twelve parameters per rate type deliver rates that rise smoothly from a (possibly negative) minimum to an asymptotic maximum in each income dimension, and are estimated separately for ETR, $MTRx$, and $MTRy$ (DeBacker et al., 2019; DeBacker and Evans, 2024a).

The framework also supports simpler forms on total income $x + y$: the three-parameter Gouveia and Strauss (1994) schedule

$$\tau^{etr}(x+y) \;=\; \phi_0\Big[1 - \big((x+y)^{-\phi_1} + \phi_2\big)^{-1/\phi_1}\,(x+y)^{-1}\Big], \tag{15}$$

whose marginal-rate schedule is the analytical derivative of the same fitted parameters — so effective and marginal rates are mutually consistent by construction — and the log-linear progressivity form of Heathcote et al. (2017), $\tau = 1-\phi_0(x+y)^{-\phi_1}$. The OG-UK wiring deployed in PolicyEngine Macro fits the Gouveia–Strauss form (15) to the PolicyEngine effective-rate microdata for each year of a three-year budget window, taking marginal-rate schedules as its analytical derivatives; the richer DEP form (13)–(14) remains available through OG-Core's `tax_func_type` option at the cost of a heavier estimation step.

4.3 Age pooling

Because ability profiles $e_{j,s}$ vary by age, tax functions can be estimated per age. The integration exposes three settings: `pooled` (one function for all ages — fastest and most stable, and the CLI default), `brackets` (one per age bracket, four groups split at the state pension age), and `each` (one per single year of age, $S = 80$ functions — closest to DeBacker et al. (2019)'s baseline but slowest and most fragile in thin cells). Pooling trades away age variation in effective schedules — pension-age interactions in particular — for robustness of the fit and of the equilibrium solve.

5 Implementation and integration

5.1 Software architecture

The stack has three layers, all open-source Python. At the bottom, **OG-Core** (`PSLmodels/OG-Core`) holds the country-agnostic theory of Section 3: the parameter apparatus, the `txfunc` estimation module, and the steady-state and TPI solvers, with transition-path lifetime problems distributed across Dask workers. Above it, **OG-UK** (`PSLmodels/OG-UK`, package `oguk`) supplies the UK calibration — demographics, macro parameters, and the PolicyEngine microsimulation hookup — and a deliberately small scoring API: `solve_steady_state(start_year, policy, ...)`, `run_transition_path(...)`, and `map_to_real_world(baseline, reform)`, which converts model-unit changes into current-price £bn using a GDP-anchored scale factor with levels tied to live ONS series (GDP, consumption, investment, government spending, the debt ratio) and HMRC total receipts, falling back to cached values offline. At the top, **policyengine-macro** (the PolicyEngine Macro integration package) wraps `oguk` in the suite's common adapters.

A reform is the suite's shared object — a flat {`parameter_path: value`} dictionary over PolicyEngine UK parameters, e.g. {`"gov.hmrc.income_tax.rates.uk[0].rate": 0.21`} for a 21% basic rate. The adapter builds a PolicyEngine `Policy` object from it, solves baseline and reform steady states, and returns levels, £bn changes, percent changes, and baseline/reform interest rates, together with the suite's common `ScoreResult` block (model class `olg-ge`, horizon `steady-state`) so that the same reform can be laid side by side with the OBR-emulator and microsimulation scores. Structural shocks outside statute — corporation tax (`cit_rate`), TFP (Z), demographics, spending shares — enter through `param_overrides` passed to the solvers, and can be combined with a statutory `Policy` in a single run.

5.2 The CLI: `pe-macro score --model og`

The command-line surface is two commands (plus the unified entry point):

```
pe-macro og-baseline
pe-macro og-score --reform
   '{"gov.hmrc.income_tax.rates.uk[0].rate": 0.21}'
pe-macro score --country uk --model og --reform '...'
```

Both use the fast configuration: pooled-age tax functions, a single representative sector, and steady-state comparison only. The baseline solve is cached at module level per `(start_year, max_iter)`, so repeated reform scores within a process pay for one baseline. The OG member is UK-only in the scorer today; passing another country is rejected with an explicit error.

5.3 Why local-only

The model is deliberately excluded from the hosted PolicyEngine Macro MCP server. A single steady-state solve — PolicyEngine microdata calibration plus the general-equilibrium fixed point — has been measured at over seventeen minutes on a laptop, a reform score needs two solves, and a transition path takes hours. That is incompatible with an interactive hosted tool whose other members answer in seconds, so the OG member runs only locally, and the suite's documentation states the runtime expectation up front rather than letting a hosted request time out.

5.4 Pinned-version fragility

The integration pins `oguk==0.3.0`, which itself pins `policyengine-uk==2.88.0`. This is a real fragility, stated plainly: `policyengine-uk` $\geq$ 2.89 renamed the microdata dataset keys the calibration downloads (from `enhanced_frs_2023_24_<year>` to `populace_uk_*`), so mixing the pinned `oguk` with a newer `policyengine-uk` fails with a `KeyError` at calibration time. The adapter traps this failure mode and raises an actionable error naming both causes — the version mismatch, and the other common environment failure, a missing `HUGGING_FACE_TOKEN` for the gated enhanced-FRS dataset — rather than surfacing a bare stack trace. Until the pin is lifted upstream, the environment that runs OG-UK must hold `policyengine-uk==2.88.0`, which also means the OG member scores reforms against a slightly older statute vintage than the suite's microsimulation member, which tracks the latest `policyengine` release. This divergence is a known, declared inconsistency of the current wiring, not a hidden one.

6 Calibration

OG-UK's calibration strategy follows the OG-Core template (DeBacker and Evans, 2024a; DeBacker et al., 2025): demographics from international and national statistical sources, macro parameters anchored to UK national accounts and official forecasts, preference parameters from the literature or matched to observed aggregates, and the tax detail supplied entirely by the estimated tax functions of Section 4. Table 1 collects the headline values used by the deployed configuration.

Three remarks on the table. First, the lifetime-ability apparatus (J ability types, shares λ_j, and profiles $e_{j,s}$) is inherited from the OG-Core default calibration, which estimates ability profiles from tax microdata for the United States; a UK-specific re-estimation of $e_{j,s}$ from the

Table 1: Headline calibration of the deployed OG-UK configuration

Parameter	Symbol	Value	Source / target
Active life span	S	80 (ages 20–99)	model structure
Childhood periods	E	20 (ages 0–19)	model structure
Population by age, mortality, migration	$\omega_{s,t}, \rho_s$	UK profiles	UN World Population Prospects (United Nations, Department of Economic and Social Affairs, Population Division, 2024)
Coefficient of relative risk aversion	σ	1.5	OG-Core default
Frisch elasticity of labour supply	θ	0.35	Blundell et al. (2007)
Discount factor	β	matched	ONS household saving ratio
Depreciation rate	δ	0.065 / yr	ONS capital stocks and consumption of fixed capital
Long-run productivity growth	g_y	0.011 / yr	OBR potential-output growth (Office for Budget Responsibility, 2025a)
World interest rate (open-economy features)	r^*	gilt-based	Bank of England yield data
Corporate tax rate	τ^b	0.27 effective	UK corporation-tax schedule (structural override)
Long-run debt-to-GDP target	α_D	ONS-anchored	fiscal closure rule, Section 3.3
Tax functions	$\tau^{etr}, \tau^{mtrx}, \tau^{mtry}$	estimated	PolicyEngine UK on enhanced FRS (Section 4)
Production sectors	M	1 (option: 8)	ONS Blue Book supply-and-use tables when $M = 8$
Tax-function age treatment	—	pooled	CLI default; `brackets`/`each` available

Table 2: Calibration targets and official UK counterparts

Quantity	Official UK value	Model treatment
Labour share of income	$\approx$ 0.59–0.60 (ONS unadjusted share)	$1 - \gamma$ set from national accounts factor shares
Capital–output ratio	$\approx$ 3 (ONS net capital stock / GDP)	emerges from $(\gamma, \delta, \beta, g_y)$; checked, not imposed
Depreciation / capital	$\approx$ 6–7% (ONS CFC / net stock)	imposed, $\delta = 0.065$
Government debt-to-GDP	$\approx$ 96–100% (PSND ex BoE)	imposed via closure target α_D; levels re-anchored to live ONS debt series in `map_to_real_world`
Potential growth	1.1% (OBR EFO)	imposed, $g_y = 0.011$
Household saving ratio	ONS series	targeted through β

enhanced FRS is an open calibration task, and results that depend on the upper tail of the ability distribution should be read with that in mind. Second, the discount factor is not taken from the literature but matched so that the model's steady-state saving behaviour is consistent with the ONS household saving ratio, given the other parameters. Third, the multi-sector option ($M = 8$ industries calibrated from ONS Blue Book supply-and-use tables by SIC section) exists for sector-level questions such as energy-price shocks, but the deployed scoring configuration is single-sector.

6.1 Model targets against official UK aggregates

Because the £bn mapping is GDP-anchored (Section 5.1), the levels the model reports are anchored to ONS series by construction; the informative comparison is between the ratios the calibration *targets* and the corresponding official numbers. Table 2 reports the comparison for the ratios the calibration speaks to directly. Where the deployed configuration anchors a quantity to the official series by construction, the table says so rather than presenting the match as an independent validation — an honesty rule this suite applies throughout.

The distinction in the final column matters for interpretation. Imposed quantities (depreciation, growth, the debt target) cannot disagree with the data; the genuinely over-identifying checks are the ones the steady state must *deliver* — the capital–output ratio and the interest rate consistent with the calibrated (γ, δ, β) triple. In the deployed fast configuration these emerge in an economically sensible range (a baseline steady-state real interest rate in low single digits, reported by `og-baseline` alongside the model-unit aggregates), but no systematic published reconciliation of achieved-versus-target moments exists for OG-UK 0.3.0, and this paper does not manufacture one: producing that reconciliation table from logged solves is the single most valuable next step for this member (Section 7).

7 Limitations

This section states plainly what the psl-og member is not, in the spirit of the suite-wide rule that a model's status — replication, emulation, or structural counterfactual — is part of its result.

No ground truth. The OBR emulator and the SVAR members of the suite are validated against published figures; the OLG member has nothing comparable to replicate. Its outputs are structural counterfactuals: internally consistent general-equilibrium answers *conditional on* the model's preferences, technology, demographics, and closure rule. Two defensible OLG models can disagree materially on the long-run effect of the same reform — the OBR's own UK OLG model (Office for Budget Responsibility, 2025b) provides a natural cross-check, and side-by-side comparison against it is the closest available substitute for validation — so point estimates should be read as model-consistent magnitudes and signs, not forecasts.

Experimental-grade wiring. The integration is young. The version-pin fragility of Section 5.4 means the OG member currently runs against `policyengine-uk` 2.88.0 while the suite's microsimulation member tracks the latest release, so the two members can disagree about baseline statute; the gated enhanced-FRS microdata adds an access prerequisite; and the CI cannot exercise a seventeen-minute solve on every commit, so regressions in solve behaviour are caught by scheduled runs and by hand, not continuously.

Steady-state-only scoring today. The CLI compares long-run steady states. That is not a budget-window costing: it answers "where does the economy settle?", not "what happens in each of the next five fiscal years?". The transition-path solver exists and is reachable through the `oguk` API, but it is hours-long, Dask-distributed, and not yet wired into `pe-macro score`. Every scoring payload carries this caveat explicitly in its `ScoreResult` block.

Fast-configuration compromises. The deployed configuration pools tax functions across ages and uses one production sector. Pooling smooths over pension-age discontinuities in effective schedules; a single sector rules out compositional questions. Both dials can be turned up (`brackets/each`, `multi_sector=True`) at material runtime and stability cost — OLG solves can fail to converge on extreme reforms, and the documented mitigation (raise `max_iter`, fall back to pooled) is a workaround, not a guarantee.

Calibration vintage and inheritance. Calibration inputs carry vintages: demographic profiles from a UN release, depreciation and factor shares from a Blue Book vintage, potential growth from a particular EFO, and — most importantly — lifetime-ability profiles $e_{j,s}$ inherited from the US-estimated OG-Core defaults rather than re-estimated on UK microdata. No published achieved-versus-target moment reconciliation exists for the deployed version. Results that lean on the upper tail of the earnings distribution, or on fine age detail around retirement, are the least trustworthy outputs of the current calibration.

Model-class assumptions. Finally, the standard structural caveats apply: perfect foresight (no aggregate risk), rational optimising households, no involuntary unemployment, exogenous productivity growth, and a closure rule that mechanically stabilises debt. These are the assumptions that buy general-equilibrium discipline; conclusions that would reverse under plausible relaxations of them should not be attributed to the data.

8 Conclusion

The psl-og member gives the PolicyEngine Macro suite something none of its other members can: a general-equilibrium answer to long-run structural questions, in which a statutory reform changes wages, interest rates, saving, and labour supply through the optimising behaviour of eighty age cohorts. The model itself is community infrastructure — OG-Core (DeBacker and Evans, 2024a) and its OG-UK calibration (DeBacker et al., 2025) — and the suite's contribution is the wiring: the same flat PolicyEngine reform dictionary that drives the microsimulation and the OBR bridge also drives a full OLG solve, with reforms entering as tax functions estimated from PolicyEngine UK microdata in the manner of DeBacker et al. (2019), and results emerging in current-price £bn under a common, comparable score schema.

The member's constraints are structural and declared: it is local-only because a solve takes tens of minutes; it is steady-state-only in the CLI today; it rests on a pinned and slightly stale dependency pair; and it has no ground truth, so its numbers are counterfactuals, not forecasts. The near-term agenda follows directly: lift the `policyengine-uk` pin so all suite members share a statute vintage; publish an achieved-versus-target calibration reconciliation for the deployed version; re-estimate lifetime-ability profiles on UK microdata; wire the transition path into the scorer so the model can speak to fiscal-year profiles; and stand up a systematic comparison against the OBR's UK OLG model (Office for Budget Responsibility, 2025b) as the closest available external check. Each of these moves the member from experimental-grade wiring towards the standard the rest of the suite is held to; none changes what it fundamentally is — an open, inspectable structural model whose assumptions are on the table.

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